

## Short Review of Conditional Probability

### Bayes' Theorem

#### Proof

Let  $A$  and  $B$  be events with  $\mathbb{P}(\cdot) > 0$ . By definition of conditional probability,

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad \mathbb{P}(B | A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}.$$

Thus,  $\mathbb{P}(A \cap B) = \mathbb{P}(A | B) \mathbb{P}(B) = \mathbb{P}(B | A) \mathbb{P}(A)$ .

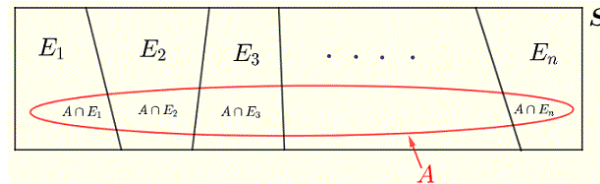
Substituting this into the first expression gives

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A) \mathbb{P}(A)}{\mathbb{P}(B)}.$$

#### Generalizing

From the law of total probability:

$$P(A) = P(A \cap E_1) + P(A \cap E_2) + P(A \cap E_3) + \cdots + P(A \cap E_n) = \sum_{i=1}^n P(A | E_i) P(E_i)$$



Using the definition of conditional probability:

$$P(A) P(E_i | A) = P(E_i) P(A | E_i) \implies P(E_i | A) = \frac{P(A | E_i) P(E_i)}{P(A)}$$

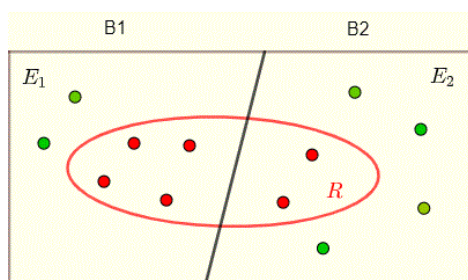
Substituting  $P(A)$  from above gives Bayes' Theorem in its general form:

$$P(E_i | A) = \frac{P(A | E_i) P(E_i)}{\sum_{j=1}^n P(A | E_j) P(E_j)}$$

## Examples (with solutions)

**Example 1.** Two boxes:

- Box 1 contains 4 red and 2 green balls.
  - Box 2 contains 4 green and 2 red balls.
  - Probability of choosing Box 1 is  $1/3$ ; Box 2 is  $2/3$ .
  - A box is selected at random and a ball is drawn.
- a) Given the ball is red, find the probability it came from Box 1.
- b) Given the ball is red, find the probability it came from Box 2.
- c) Compare and explain the results.



**Solution.**

Let:  $E_1 = \{\text{Box 1 is selected}\}$ ,  $E_2 = \{\text{Box 2 is selected}\}$ ,  $R = \{\text{a red ball is selected}\}$

We have:  $P(E_1) = \frac{1}{3}$ ,  $P(E_2) = \frac{2}{3}$ ;  $P(R | E_1) = \frac{4}{6} = \frac{2}{3}$ ,  $P(R | E_2) = \frac{2}{6} = \frac{1}{3}$

$$\text{a) } P(E_1 | R) = \frac{P(R|E_1) P(E_1)}{P(R|E_1) P(E_1) + P(R|E_2) P(E_2)} = \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3}} = \frac{1}{2}$$

$$\text{b) } P(E_2 | R) = \frac{P(R|E_2) P(E_2)}{P(R|E_1) P(E_1) + P(R|E_2) P(E_2)} = \frac{\frac{1}{3} \cdot \frac{2}{3}}{\frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3}} = \frac{1}{2}$$

- c) The probabilities are equal. Although Box 1 has twice as many red balls as Box 2, Box 2 is twice as likely to be chosen. Bayes' Theorem accounts for both effects.

**Example 2.** Disease testing:

- 1% of the population has the disease (event  $D$ ); the remaining 99% do not (event  $\neg D$ ).
- The test is positive in 95% of those with the disease, and falsely positive in 2% of those without.
- A randomly chosen person tests positive ( $TP$ ). What is  $P(D | TP)$ ?

**Solution.**

$$P(D | TP) = \frac{P(TP | D) P(D)}{P(TP | D) P(D) + P(TP | \neg D) P(\neg D)} = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.02 \cdot 0.99} \approx 0.32.$$

Despite a positive test, the probability of having the disease remains low bc the “disease-free” population is so much larger. Using a hypothetical cohort of 1000 people:

$$990 \text{ disease-free people: } 0.02 \cdot 990 = 19.8 \approx 20 \text{ false positives;}$$

| $D$ | $FD$ |
|-----|------|
| 1%  | 99%  |
| 95% | 2%   |

10 with disease:  $0.95 \cdot 10 = 9.5 \approx 10$  true positives.

Total positives =  $20 + 10 = 30$ , of which only 10 truly have the disease:  $P(D | TP) = \frac{10}{30} \approx 0.32$ .

**Example 3.** Defective bulbs:

- Three factories A, B, C produce 20%, 50%, 30% of bulbs, respectively.
- Defect rates are: A — 2%, B — 1%, C — 3%.
- A randomly selected bulb is defective ( $D$ ). What is the probability it came from factory B?

**Solution.**

$$P(B | D) = \frac{P(D | B) P(B)}{P(D | A) P(A) + P(D | B) P(B) + P(D | C) P(C)} = \frac{0.01 \cdot 0.50}{0.02 \cdot 0.20 + 0.01 \cdot 0.50 + 0.03 \cdot 0.30} \approx 0.2777.$$

Even though B produces half the bulbs, its low defect rate means a defective bulb is less likely to have originated there.

**Example 4.** Radar detection:

- An aircraft is present ( $A$ ) with probability 7%; no aircraft ( $A^c$ ) with 93%.
- The radar correctly detects an aircraft with probability 98%; falsely detects when none is present with probability 5%.

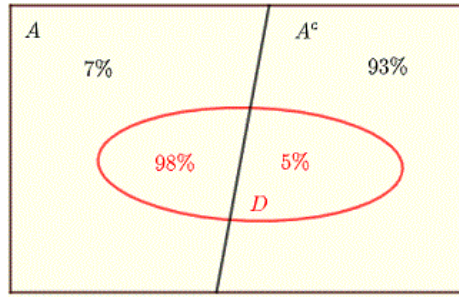
Find:

- $P(A^c | D)$ : aircraft detected ( $D$ ), but none present.
- $P(A | D)$ : aircraft detected and present.
- $P(A | D^c)$ : aircraft not detected but present.
- $P(A^c | D^c)$ : no detection and none present.

**Solution.**

We have

$$P(A) = 0.07, \quad P(A^c) = 0.93, \quad P(D | A) = 0.98, \quad P(D | A^c) = 0.05.$$

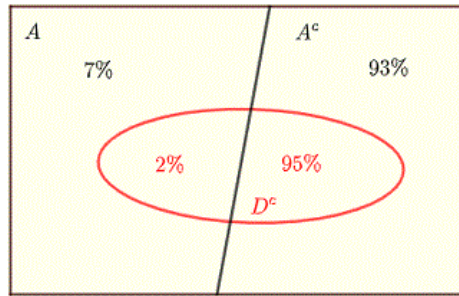


$$a) P(A^c | D) = \frac{P(D|A^c) P(A^c)}{P(D|A^c) P(A^c) + P(D|A) P(A)} = \frac{0.05 \cdot 0.93}{0.05 \cdot 0.93 + 0.98 \cdot 0.07} \approx 0.4040.$$

$$b) P(A | D) = \frac{0.98 \cdot 0.07}{0.98 \cdot 0.07 + 0.05 \cdot 0.93} \approx 0.5960.$$

$$c) P(D^c | A) = 0.02, \quad P(D^c | A^c) = 0.95.$$

$$\text{Then } P(A | D^c) = \frac{0.02 \cdot 0.07}{0.02 \cdot 0.07 + 0.95 \cdot 0.93} \approx 0.0016.$$



$$d) P(A^c | D^c) = \frac{0.95 \cdot 0.93}{0.95 \cdot 0.93 + 0.02 \cdot 0.07} \approx 0.9984.$$

